How the tool thinks

Five ideas explain almost everything the tool does.
The rest of the guide leans on them.

Ticks. Lines, as on a minute track.

Markers. Bars, wedges or dots, usually for the hours.

Numerals. Numbers or words, in any font.

Band. A flat ring of colour, whole or in sectors.

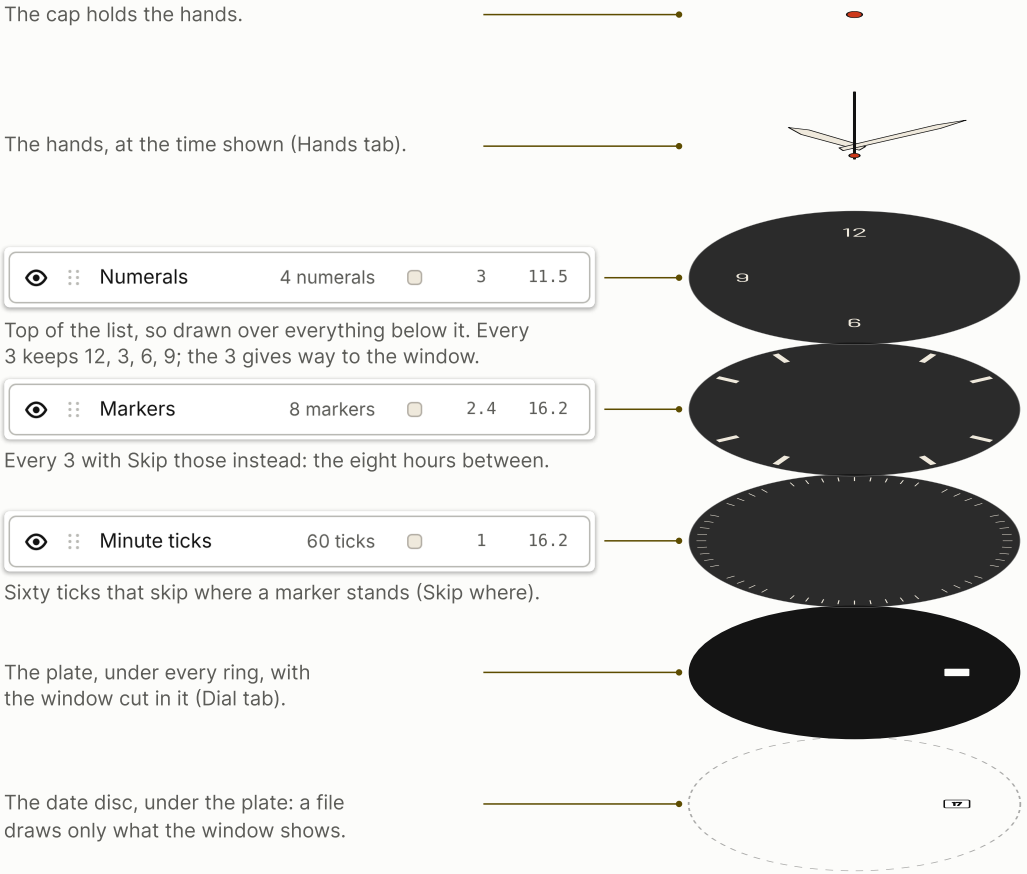


The First dial (page 05), as the tool draws it.

The panel edits what is selected. Select a ring in the list, or tap one of its marks on the dial: everything below the list now edits that ring. The hand tabs in Hands and the window cards in Windows work in just the same way.

Cap, hands, print, plate, disc. A watch dial, seen from the side, is a short stack: from the top, the cap that holds the hands, the hands, the plate with its print, and the date disc under the plate. The tool draws that stack from the bottom up. The print on the plate is made of rings.

Exploded, not to scale, top to bottom. A row's two numbers are its size and radius.



A ring repeats one thing round the centre. A ring is ticks, markers, numerals or a band, repeated at its positions: **Count** of them, from **Start °**, across **Span °**. **Every** keeps one position in so many; **Skip those instead** keeps the others. Every 3 on twelve positions keeps 12, 3, 6, 9; with Skip those instead, the eight hours in between them.

True size. Every length is in millimetres, measured on the dial; files come out at true size. The page is the diameter plus twice the page margin: a 30 mm dial with a 3 mm margin makes a 36 mm page, and the file is still named after the dial.

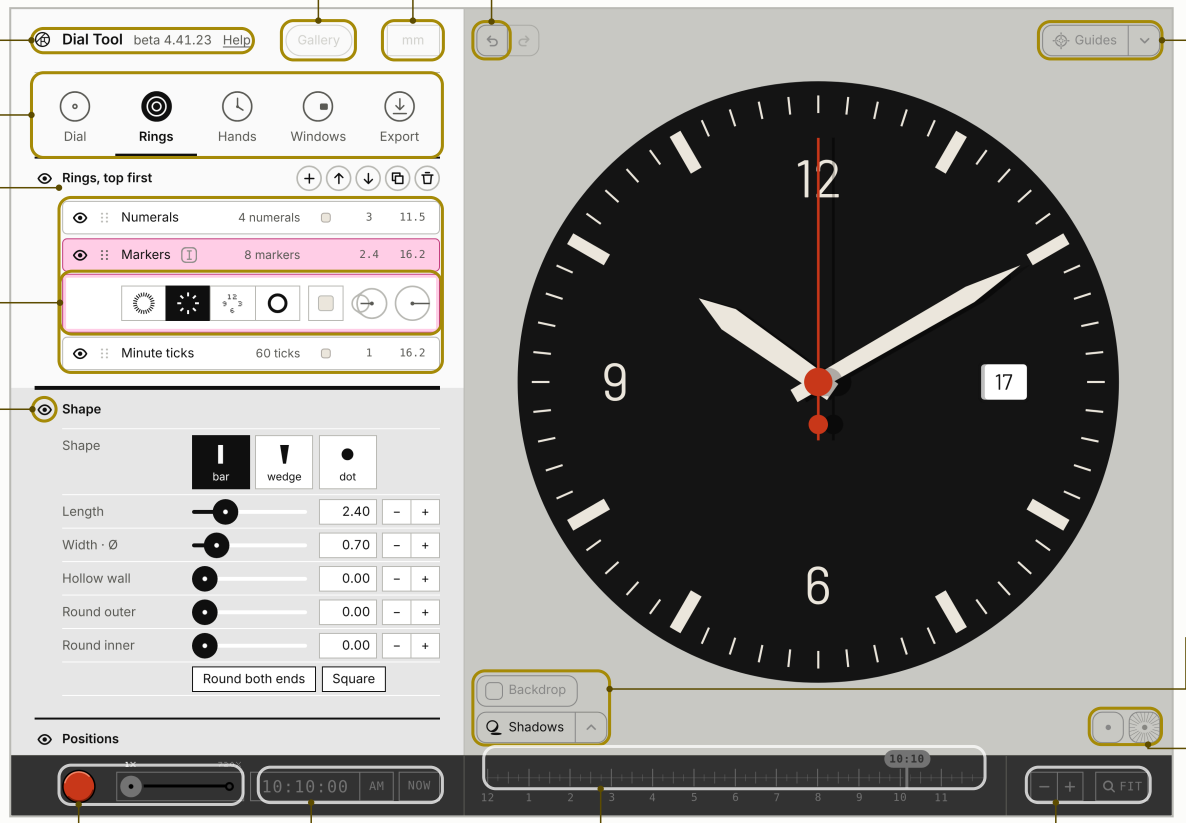
The higher ring wins. A dial is a list of rings, top first. A ring below gives way to the rings above in one of two ways, under Rings › Give way to rings above. **Skip where** leaves out its own marks at the angles where a ring above has one: minute ticks that skip where markers stand. **Knock out** keeps its marks but cuts them round the rings above, leaving a gap the width of the Clearance.

Kept as you go. There is no Save to press. About half a second after you stop, the dial is kept in this browser and written into the page's address. The address is a link to the dial: send it and the dial opens at the time shown, though not on your tab or in your unit.

⚠ **Watch.** While time is playing nothing is kept; the dial is kept once you stop it.

The screen

The panel on the left, the dial on the right, the bottom bar along the foot.
Pictures from an iPad held landscape; a Mac is the same with more room.



The unit numbers are shown and typed in: mm, points, inches or lignes. The dial stays in millimetres.

Gallery: dials you can open and take apart.

Undo and redo: step back and forth through your changes.

The mark opens a note about the tool; Help opens this guide.

Five tabs choose what the panel edits; the dial and the time stay as they are.

The rings, top first, and buttons to add, move, duplicate or delete one. Everything under the list edits the selected ring.

The drawer of the selected ring: kind, colour, size and radius (page 03).

The eye beside a pane's name folds it to one line, and opens it again.

Guides: construction lines, never exported. Press to hide them; the arrow picks which.

Typing numbers. A number field takes sums: 1.2 sets, +0.5 adds, x0.5 multiplies, /2 divides.

Gallery and mm. Both are drawn light, but they are live: a press works.

Backdrop and Shadows: how the dial is seen, never exported.

The dial alone: the panel slides away. Beside it, the showcase: a dark view; press again to leave.

Play, and its speed from 1× to 720×. While time plays, nothing is kept.

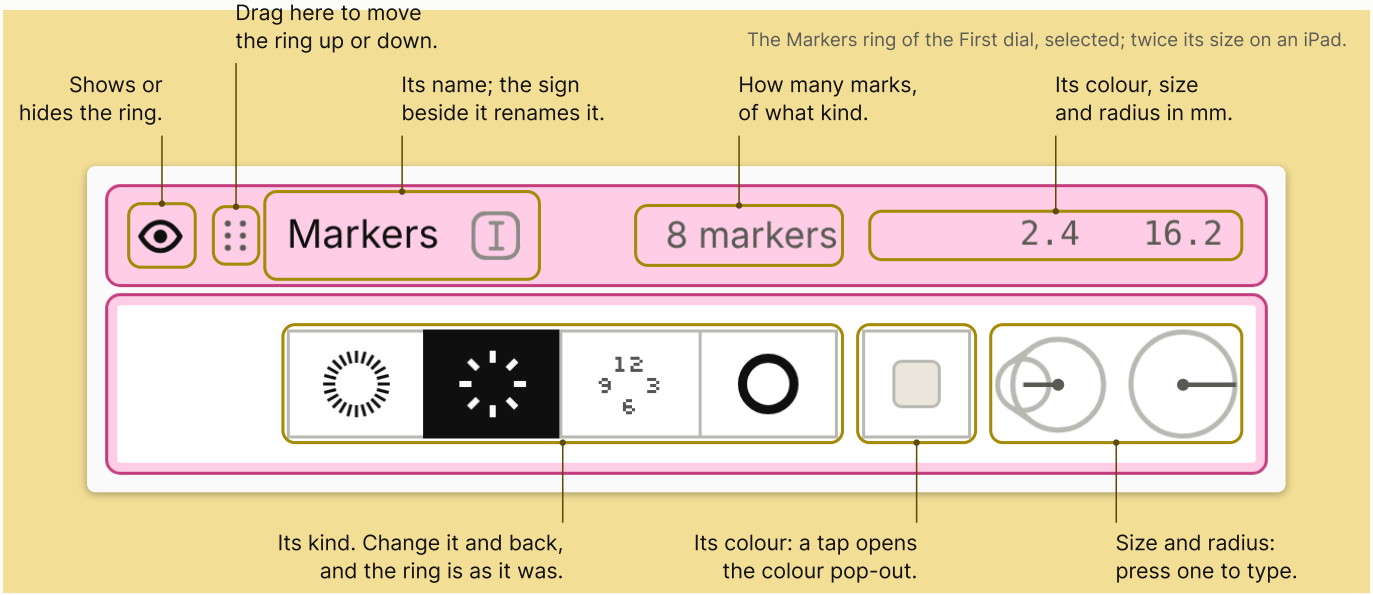
The time: type it, or press NOW for the time it is.

Drag across twelve hours; its steps are the hours and 10:10, the classic pose.

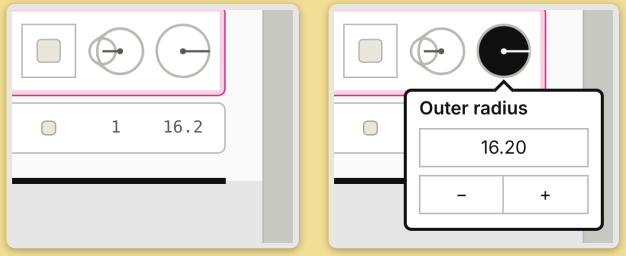
Zoom, and FIT to see the whole dial.

A ring, close up

Each ring has a row in the list, and a drawer under the row when it is selected. Most of what you change is in the drawer and the panes under it.



The Markers ring of the First dial, selected; twice its size on an iPad.

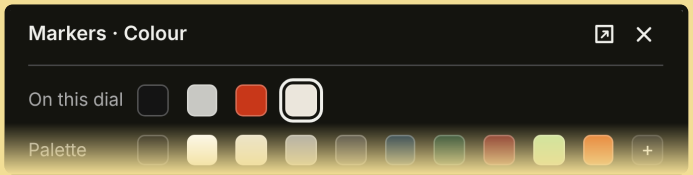


The radius button, pressed: its pop-up.

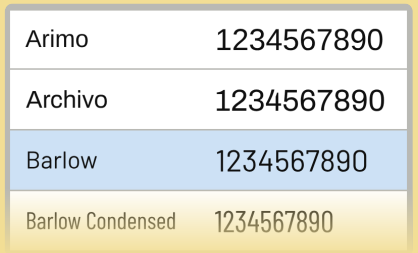
The round buttons. Press one and its number opens in a small pop-up: on a Mac it is ready to type over; on a touch screen, tap the number first. A tap anywhere else on the screen closes it.

The ring is	First button	Second button
ticks, markers	Length: of one mark, inward from its outer end	Outer radius: where the marks' outer ends sit
numerals	Size: the type size in mm, as in any layout program	Centre radius: where the numerals' centres sit
band	Band width	Outer radius

⚠ **Watch.** A numerals ring's radius can measure to the numerals' centres, their baseline or their tops: the small menu in the radius button's pop-up. Centre is the start.



Colours. A colour opens a pop-out beside it: its × closes it; the square beside the × makes it twice the size.



Fonts. Change… opens the font sheet, which stays open after you choose: tap outside it to close it. The fonts listed there ship with the tool and look the same everywhere.

The screen on a phone

The same tool, stacked: the bar on top, the dial, the panel as a drawer. Everything works by finger.

The bottom bar moves to the top. The round red button at its left is Play: a tap starts time, and while it runs nothing is kept, so tap it again to stop.



Drag the grey pill on the drawer's top line to make the drawer taller or shorter.

The switches over the dial show only their signs.

The title line, with Help, Gallery and mm, shrinks once you scroll.

Long ring names are cut short; the row's "4 numerals" tells them apart.

A phone held upright, 390 points wide, at 0.17 mm a point.

The font sheet on a phone. It covers the dial as well as the drawer, so a tap "outside" it can land on a font: close it with a tap on the dark bar at the top. Colour pop-outs close with their ×.

Your first dial

A black field watch, 34 mm, in ten steps; the pictures are about true size. Each picture is a link: tap it to open the dial as it is after that step.



1 · Start from the starting dial. Export › Settings file › **Defaults** (scroll the panel down to it). It replaces the dial you had; ⌘Z or the undo arrow brings it back.



2 · Size. Dial › Diameter: type 34.



3 · Colours. Dial › Plate: #141414, a near-black. Then in Colours, set Numerals, Markers, Minute ticks, Hour hand and Minute hand to #EFE9DC, a warm white: tap the colour chip beside each name, then HEX, and type.



4 · A font that ships with the tool. Rings › select Numerals › Type › **Change…**, choose Barlow, then tap outside the sheet. The starting dial asks for Helvetica Neue; where it isn't installed, the panel says "not loaded" and Inter stands in.



5 · Numerals. With Numerals selected: its radius button, 11.5; its size button, 3. Under Positions, Every: 3. Now only 12, 3, 6 and 9 are left.



6 · Markers and ticks. Select Markers: radius 16.2, length 2.4; Every 3 and **Skip those instead**, so the bars stand at the eight hours between. Select Minute ticks: radius 16.2. They already skip where markers stand.



7 · The date at 3. Windows › the card at 3 › **Show this window**. The numeral at 3 gives way to it: a window that is shown leaves out the numeral and the marker at its position, at any radius.



8 · Hands. Hands › Hour › shape **sword**; then Minute › **sword**. Length runs from the centre to the tip, Tail behind the centre, Width across the blade. The seconds hand and the cap keep their starting red.

Your first dial, continued

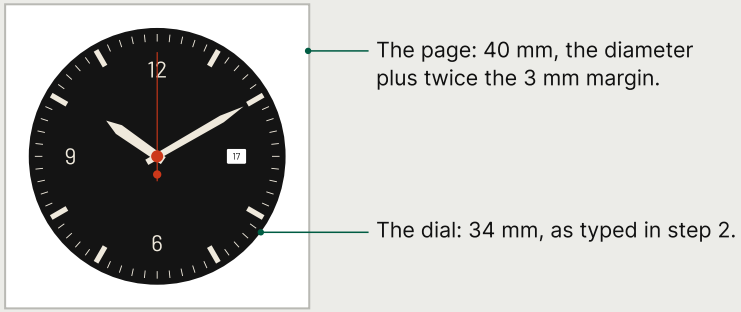
The pose and a file, and what the ten steps make.
The file is shown here at its true size.



9 · The pose. In the bottom bar, the 10:10 step.



10 · A file. Export › **PDF**. It opens at true size: a 34 mm dial on a 40 mm page. Before you print from it, read For the maker, on page 08.



The PDF of step 10, printed at 100 %: true size.



The First dial: what the ten steps make.

Find it

Questions readers asked, and where the answer is.
Each answer was checked on the version named at the top.

I want to start fresh. Export › Settings file › Defaults, or any dial in the Gallery. Either replaces your dial; ⌘Z brings it back. There is no empty dial yet.

Where do colours live? Every colour on the dial is in one list: Dial › Colours, a chip beside each name. A ring's own colour is also in its drawer; a hand's colour is in its pane.

Why did the 3 disappear? A window that is shown takes its place: the numeral and the marker at its position are left out, at any radius. If a minute tick was hidden under that marker, it shows again. There is no way to keep the numeral yet.

Why isn't my window showing? The four windows exist from the start but are hidden: Windows › its card › Show this window. A card says "(not shown)" in its name until then. Width and Height size the hole; Size % sizes the date in it, as a share of the hole's height.

Window at 6, or Window 3? The cards name a window by where it is; the colour list and the file's layers number them in the order they came: Window 1 is the card at 3, then 12, 6 and 9.

I asked for four numerals and got 12, 1, 2, 3. Numerals follow the positions in order. For 12, 3, 6, 9 keep Count at 12 and set Every to 3.

Skip where, or Every? Every chooses a ring's positions by counting, whatever is above it. Skip where leaves out positions where a ring above has a mark, and follows that ring when it changes. Minute ticks under markers: Skip where. Bars between the numerals: Every, and tick Skip those instead.

The panel says "not loaded". The font isn't on this device. Inter stands in until you choose one with Change…; the fonts listed there ship with the tool and look the same everywhere.

My time isn't 10:10. The time keeps its seconds. Press the 10:10 step in the bottom bar, or type 10:10:00.

Where did my dial go? It is kept in this browser and in the address. Another browser or device doesn't have it: send yourself the link (Export › Link to this dial › Copy link), or save a settings file (Export › Settings file › Save…).

How do I leave the showcase? Press its switch again, bottom right.

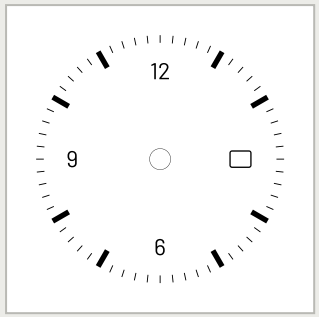
A number field takes sums. Type 1.2 to set a value, +0.5 to add, x0.5 to multiply and /2 to divide.

The signs without words.

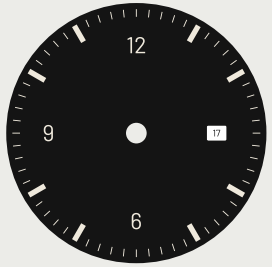
Sign	Where	What it does
Eye	each ring's row	shows or hides the ring
Grip (six dots)	each ring's row	drag to reorder
Text cursor in a box	the selected row	renames the ring
+ ↑ ↓ and two more	the ring list's heading	add a ring · move up · move down · duplicate · delete
Four tiles	the ring drawer	the ring's kind: ticks, markers, numerals, band
Two round buttons	the ring drawer	size · radius: press to type
Four small dials	Windows	the windows whose cards sit at 12, 3, 6, 9
Square with rays	over the dial, bottom right	the dial alone
Square with a dot	over the dial, bottom right	the showcase
Arrows	over the dial, top left	undo · redo

For the maker

One page for whoever etches or prints the dial.
File facts only; advice on each process waits for a first etch and a shop's questions.



The First dial as the print mask: the toner or resist.



The same dial as a colour file, hands hidden.

What a file holds. A colour export holds the plate, every ring, the windows with their content, the hands at the time shown, and the cap. Guides, the backdrop and shadows are never in a file. Hide the hands (Hands › Show hands) before exporting a colour file for print: hidden hands are left out.

Size. True size, in millimetres. The page is the diameter plus twice the margin; the file name gives the diameter. A PNG has its dpi written in (Export, the number under PNG), so it opens at true size. The Export log names each file and its size.

The print mask. Export › Print › Mono print mask: every ring and the windows' frames in black on white, without plate, hands or window content. The edge of each window's hole and the centre hole are 0.05 mm cut lines. The dial on screen shows the mask while it is on.

Mirrored, for toner transfer only. It flips the file left to right. Never send a mirrored file to a pad or screen printer.

⚠ **Watch.** Marks lying on a band disappear in the mask, black on black, unless the band knocks them out (Rings › Give way to rings above › Knock out).

⚠ **Watch.** The cut lines are in the same black as the marks. Delete the layers "Window 1 cut line" and "Hole" from the PDF or SVG before printing a resist; the PNG has them baked in. Without "Hole" there is no centre mark; the tool has no registration marks yet.

One film per ink. Export one PDF with everything shown and use its layers: one per ring, named as in the list. Don't hide rings and export again: a ring only knocks out round rings that are shown.

Colour. Colours are plain RGB values, with no colour profile. Give the printer the hex codes from Dial › Colours and a physical colour reference.

Text. Numerals and dates are outlines in every file. Export › "Keep numerals as text in the SVG" writes them as text in the SVG only; the font must then be installed where the file is opened.

The thinnest line. For ticks and markers it is in the ring's settings, in its Shape pane: Width, and Inner width × if the marks taper. Numerals have no width setting: zoom into the PDF and measure the thinnest stroke yourself; the tool doesn't report it yet.

Process advice. Minimum line, clearance, ink order: pending a first etch and a shop's questions.